

Wen Perng

 wenperng |  wenperng.github.io |  414waynep@gmail.com

EDUCATION

National Taiwan University (NTU)

Taiwan

B. S. in Engineering, Department of Electrical Engineering

Sep. 2021 – Jan. 2026

- Overall GPA: 4.23/4.3, rank: 7/268
- Dean's List Award: for semesters Fall 2021 and Spring 2022.
- Presidential Award: for 2021 - 2022 academic year.

Technische Universität München (TUM)

Germany

Exchange program

Apr. 2024 – Sep. 2024

- Grade: 1.0

Taipei Fuhsing Private School

Taiwan

High school

Sep. 2018 – Jun. 2021

RESEARCH EXPERIENCE

Multimedia Processing and Communications Lab

NTU, Taiwan

Advisor: Homer H. Chen

Feb. 2023 – Jan. 2026

- Characterized visual tolerance of step-edge artifact of light field AR displays.
- Developed all-optical system for image interpolation using amplitude-modulation.

Adaptive Systems Lab

EPFL, Switzerland

Advisor: Ali H. Sayed

Jul. 2025 – Aug. 2025

- Extended diffusion stochastic learning algorithms to multi-team network games.
- Provided stability conditions and first-order mean-square-error performance to a gradient-based algorithm termed ATC-ITC.

PUBLICATIONS

Wen Perng, Po-Hung Cheng, and Homer H. Chen (2026). *Manifold Optimization on the Magnitude Torus for Fourier Phase Retrieval*. (Accepted by ICIP2026).

Wen Perng, Vladyslav Shashkov, Haoyuan Cai, and Ali H. Sayed (2026). “Diffusion Stochastic Learning Over Multi-Team Network Games”. In: *ICASSP 2026 - 2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 936–940. DOI: [10.1109/ICASSP55912.2026.11464894](https://doi.org/10.1109/ICASSP55912.2026.11464894).

Bing-Cheng Chiang, Chih-Sheng Chang, Wen Perng, et al. (2025). “Light field AR display for close-range applications: advantages, use cases, and challenges”. In: *Advances in Display Technologies XV*. Ed. by Jiun-Haw Lee and Qiong-Hua Wang. Vol. 13388. International Society for Optics and Photonics. SPIE, p. 1338808. DOI: [10.1117/12.3041498](https://doi.org/10.1117/12.3041498). URL: <https://doi.org/10.1117/12.3041498>.

Ting-Hsun Chi, Wen Perng, and Homer Chen (2023). “Perceptual Tolerance of Split-Up Effect for Near-Eye Light Field Display”. In: *2023 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, pp. 177–184. DOI: [10.1109/ISMAR59233.2023.00032](https://doi.org/10.1109/ISMAR59233.2023.00032).

AWARDS

NTUEE Undergraduate Innovation Award

Jun. 2025

First prize and popularity award.

- Topic: “All-Optical System for Super-Resolution.”

NASA Space App Challenge

Oct. 2024

Silver award in Taipei local event.

- Topic: “Seismic Detection Across the Solar System.” [Link to project.](#)
- Responsible for coming up with the scheme of detecting seismic waves via traditional signal processing methods.

NTUEE Undergraduate Innovation Award

Jun. 2023

Third prize.

- Topic: “On the Split-Up Effect of Light Field for AR Display.”

Presidential Award (Top 2% GPA)

2021 - 2022 Academic Year

Dean’s List Award (Top 5% GPA)

Fall 2021 and Spring 2022

Asian Physics Olympiad

2021

Bronze medal.

SKILLS

Languages

English (TOEFL iBT: R/L/S/W = 30/28/26/28).

May 2025

Programming

MATLAB.

Sport

Kendo (Nidan, 2nd dan).